

Sri Chaitanya IIT Academy.,India.

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ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -1(Adv-2019-P1-Model)** **Date: 05-04-2026**

Time: 09.00Am to 12.00Pm **GTA-22** **Max. Marks: 186**

05-04-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2019-P1)_GTA-22_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

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Name of the Student: _____ **H.T. NO:** | | | | | | | | |

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JEE-ADVANCED-2019-P1-Model

Time: 3:00Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 186**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	3	-1	4	12
Sec – II(Q.N : 5 – 12)	Questions with Multiple Correct Choice **+1 partial marks**	4	-1	8	32
Sec – III(Q.N : 13 – 18)	Questions with Numerical Value Answer Type	3	0	6	18
Total				**18**	**62**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 22)	Questions with Single Correct Choice	3	-1	4	12
Sec – II(Q.N : 23 – 30)	Questions with Multiple Correct Choice **+1 partial marks**	4	-1	8	32
Sec – III(Q.N : 31 – 36)	Questions with Numerical Value Answer Type	3	0	6	18
Total				**18**	**62**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 40)	Questions with Single Correct Choice	3	-1	4	12
Sec – II(Q.N : 41 – 48)	Questions with Multiple Correct Choice **+1 partial marks**	4	-1	8	32
Sec – III(Q.N : 49 – 54)	Questions with Numerical Value Answer Type	3	0	6	18
Total				**18**	**62**

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PHYSICS **Max Marks: 62**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct. **Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)**

- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If the answer is correct)
  Zero Marks: **0** (If the question is unanswered)
  Negative Marks: -1 (In all other cases)

1. Which one of the following statements is WRONG in the context of X - rays generated from a X-ray tube?

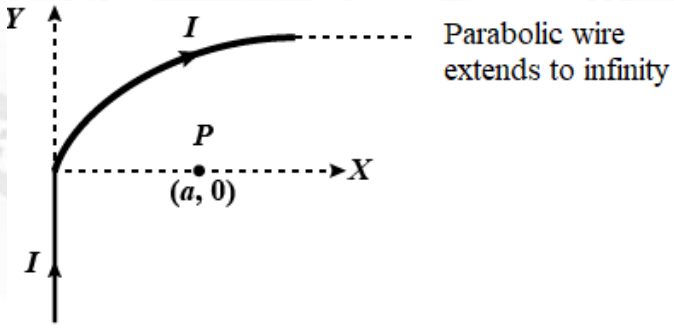
A) Wavelength of characteristic X-rays decreases when the atomic number of the target increases

B) Cut-off wavelength of the continuous X-rays depends on the atomic number of the target

C) Intensity of the characteristic X-rays depends on the electrical power given to the X-ray tube

D) Cut-off wavelength of the continuous X-rays depends on the energy of the electrons in the X-ray tube

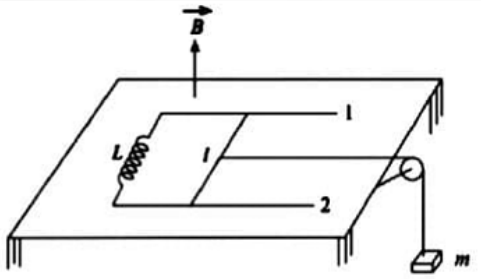
2. An infinitely long wire bent into shape as shown in figure carries a steady current I. If the curved wire is along the curve $y^2 = 4ax$, then magnetic field at point P(a ,0), is



A) $\frac{\mu_o i}{4\pi a}$ **B)** $\frac{\mu_o i}{8\pi a}$ **C)** $\frac{\mu_o i}{4a}\left(\frac{\pi+2}{2\pi}\right)$ **D)** $\frac{\mu_o i}{4\pi a}\left(\frac{\pi+4}{2}\right)$

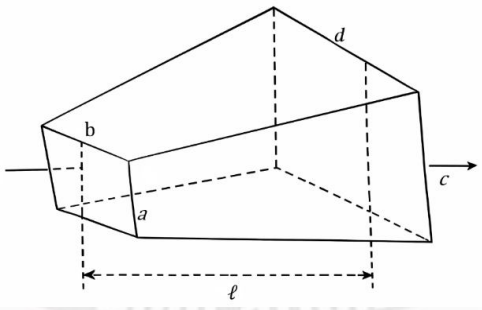
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3. A pair of fixed parallel horizontal conducting smooth rails of negligible resistance is connected to an inductance L as shown in the figure. A constant magnetic field B exists perpendicular to the table. The distance between the rail is l. A conducting rod of negligible resistance and mass *m* can slide on the rails. The rod is connected to mass *m* by a light inextensible non conducting string passing over a light frictionless pulley as shown. Calculate speed of the rod as a function of distance x from initial position.



A) $\sqrt{gx+\frac{B^2l^2x^2}{4mL}}$ **B)** $\sqrt{gx-\frac{B^2l^2x^2}{2mL}}$ **C)** $\sqrt{gx-\frac{B^2l^2x^2}{4mL}}$ **D)** $\sqrt{\frac{gx}{2}-\frac{B^2l^2x^2}{4mL}}$

4. A long prismatic conductor of varying rectangular cross-section and of length l is shown in the figure. The dimension of rectangular cross-section on the one end is $a\times b$ and on the other end it becomes $c\times d$. The side length of rectangular cross-section varies linearly with the distance from one end. The resistivity of the material of the conductor is ρ. Find the resistance of the conductor, if the current enters from one end and leaves from the other end.



A) $\frac{\rho l}{ad-bc}\ln\left(\frac{ad}{bc}\right)$ **B)** $\frac{\rho l}{a^2-b^2}\ln\left(\frac{d}{c}\right)$

C) $\frac{\rho l}{dc-ab}\ln\left(\frac{cd}{ab}\right)$ **D)** $\frac{\rho l}{d^2-b^2}\ln\left(\frac{a}{c}\right)$

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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

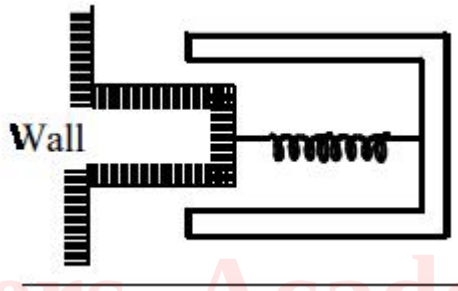
This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**

Section 2 (Max Marks: 32)

- Section two contains Eight (8) questions
- Each Question has Four Options and One Or More Than One of these four will be the correct answer.
- For each question, choose the option(s) corresponding to all the correct answer(s)

The Marking scheme to evaluate Answer to each question will be :
Full Marks: **+4** (If the answer is correct)
Partial Marks: **+3** (If all the four options are correct but **only 3** are chosen)
Partial Marks: **+2** (If THREE or More options are correct but **only 2** correct answers are chosen)
Partial Marks: **+1** (If TWO or More options are correct but **only 1** correct answer is chosen)
Zero Marks: **0** (If the question is unanswered)
Negative Marks: -1 (In all other cases)

5. Consider the situation shown in diagram in which a hollow cylindrical box of mass m, open at one end is connected to an ideal spring. Initially spring is compressed and box having mass m is kept stationary. Spring is attached with box but not attached with wall and initially held at rest. Now box is released. Choose INCORRECT options



A) Work done by wall on box plus spring system is zero

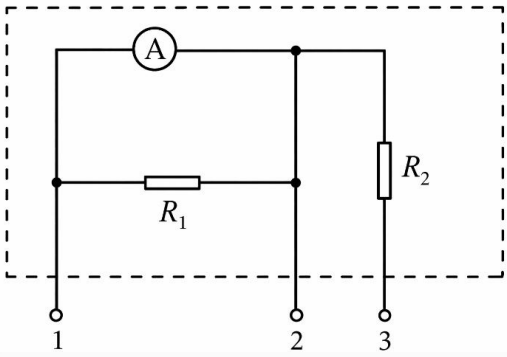
B) Work done by wall on spring plus box system is positive

C) Change in kinetic energy of box plus spring system is zero

D) Work done by wall on box plus spring system is zero but change in kinetic energy of this systems is positive, so work energy theorem fails in this case

6. As shown in the figure, the range of ammeter A is 0.6A, the dial is evenly divided into 30 small grids, each small grid represents 0.02A; the resistance R_1 is equal to the resistance of the ammeter, which is equal to 100Ω The resistance of R_2 is equal to 2 times the resistance of the ammeter.

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A) When connecting terminals 1 and 2 to the circuit, as an ammeter, each small grid represents 0.04A.

B) When connecting terminals 1 and 2 to the circuit, as an ammeter, each small grid represents 0.02A.

C) When connecting terminals 1 and 3 to the circuit, as an voltmeter, each small grid represents 10V.

D) When connecting terminals 1 and 3 to the circuit, as an voltmeter, each small grid represents 5V.

7. Figure shows a long thread along the axis of a long hollow cylinder. The uniform charge per unit length of thread is λ while uniform surface charge density of the cylinder is $-\sigma$. The radius of the cylinder is R. Then correct option among the following is/are



A) Total flux associated with a coaxial cylinder of finite length and radius twice that of the charged cylinder is zero, if the magnitude of $\sigma = \frac{\lambda}{2\pi R}$

B) Total flux associated with a coaxial cylinder of finite length and radius twice that of the charged cylinder is zero, if the magnitude of $\sigma = \frac{\lambda}{4\pi R}$

C) Total flux associated with a coaxial cylinder of finite length and radius half that of the charged cylinder is independent of σ

D) Total flux associated with a coaxial cylinder of finite length and radius half that of the charged cylinder is zero and independent of σ

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8. Suppose the potential energy between electron and proton at a distance r is given by $\frac{Ke^2}{3r^3}$, application of Bohr's theory to hydrogen atom in this case shows that

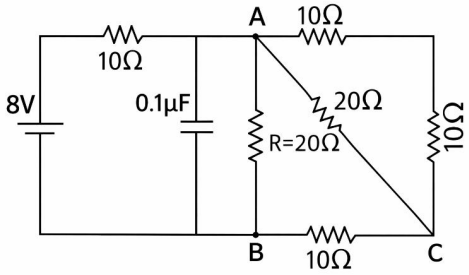
A) Energy in the nth orbit is proportional to n^6

B) Energy is proportional to m^{-3} for given 'n' (m = mass of electron)

C) Energy the nth orbit is proportional to n^{-2}

D) Energy is proportional to m^3 (m = mass of electron)

9. A capacitor of capacitance $0.1\mu F$ is connected a battery of emf 8V as shown in the fig. Under steady state condition, choose correct option(s)



A) Current in the resistor (R) between points A &B is 0.1 A

B) Energy stored on the capacitor is $0.2\mu J$

C) Charge on the capacitor is $0.4\mu C$

D) Current in the resistor (R) $20\ \Omega$ between points A and C is 0.4 A

10 For an isosceles prism of angle A and refractive index μ, it is found that the angle of minimum deviation $\delta_m = A$. Which of the following is incorrect?

A) For the angle of incidence $i_1 = A$, the ray inside the prism is parallel to the base of the prism

B) For the prism, the refractive index μ and the angle of prism A are related as $A = \frac{1}{2}\cos^{-1}\left(\frac{\mu}{2}\right)$

C) At minimum deviation, the incident angle i_1 and the refracting angle r_1 at the first refracting surface are related by $r_1 = (i_1/2)$

D) For this prism, the emergent ray at the second surface will be tangential to the surface when the angle of incidence at the first surface is $i_1 = \sin^{-1}\left[\sin A\sqrt{4\cos^2\frac{A}{2} - 1} - \cos A\right]$

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11. The current density in a long wire of radius 'a' varies with radial distance r as $J = kr^2$, where k is a constant. Then correct option(s) among the following is/are

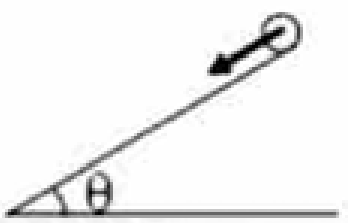
A) Total current passing through the cross section of the wire is $I = \frac{\pi k a^4}{4}$

B) Total current passing through the cross section of the wire is $I = \frac{3\pi k a^3}{2}$

C) The magnetic field at a distance $r > a$ is $B = \frac{\mu_0 k a^4}{4r}$

D) The magnetic field at a distance $r < a$ is $B = \frac{\mu_0 k r^3}{4}$

12 A solid sphere is in pure rolling motion on an inclined surface having inclination θ identify the correct option(s) among the following



A) Keeping mass and θ constant if radius of solid sphere increases, friction does not change

B) Friction do negative work with respect to incline plane

C) Torque due to friction will increase its angular velocity

D) for same θ, if μ of surface increases then pure rolling is not possible

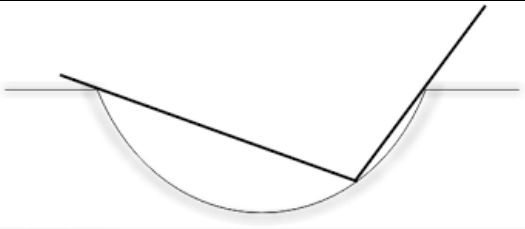
SECTION – III
(NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. **(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).**
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 3 (Max Marks: 18)

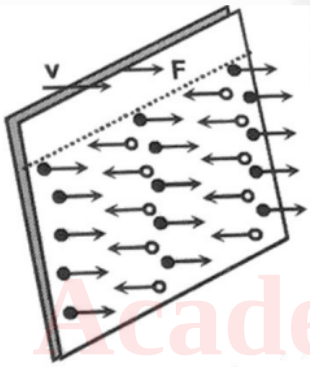
- This Section contains **Six (06)** Questions. The answer to each question is a Numerical Value
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If ONLY the correct numerical value is entered)
  Zero Marks: 0 (In all other cases)

13. A semi-cylindrical groove of radius r=20 cm is made on a horizontal floor. An ant wants to cross the groove. A boy decides to help the ant making a bridge consisting of straight wire segments. But all the wires available are of length $l = 32cm$, so the boy rigidly connects two wires at right angle and places the bridge in the groove as shown in the figure. If the ant can crawl up a wire segment at speed $\upsilon = 0.5cm/s$ and down a wire segment at speed 2υ, in what time in seconds can the ant cross the groove with the help of this bridge? The left part of the bridge is just inside the groove.

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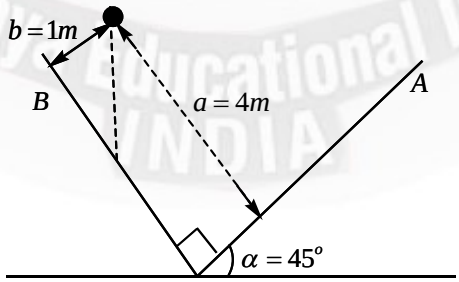


14. A thin heavy metal plate is being bombarded by a perpendicular beam of gas particles from both sides as shown in the figure. The solid dots are representing the molecules hitting from left side and the faint dots are the molecules hitting from right side. The mass of these gas particles is $m = 10^{-26} kg$ and velocity before hitting $v_0 = 5m/s$. Volume density of the gas particles on both sides is $n = 10^{25} \, per \, m^3$. Each beam has an area $A = 1m^2$ and the collisions are perfectly elastic. What is the external force F (in newton) required to move the plate with a constant velocity v = 2 m/s.



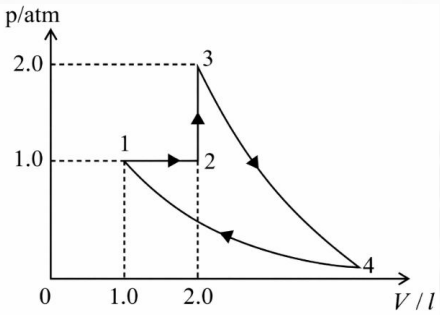
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15. Two boards have been fixed at right angles to one another. Their line of touching is horizontal. An elastic ball is released at a point at perpendicular distance a=4m from the plane A and b=1m from B. On the average, how many times does the ball bounce against wall B for each time it bounces against wall A? collisions are absolutely elastic

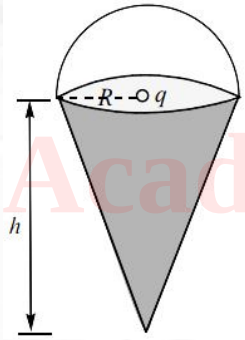


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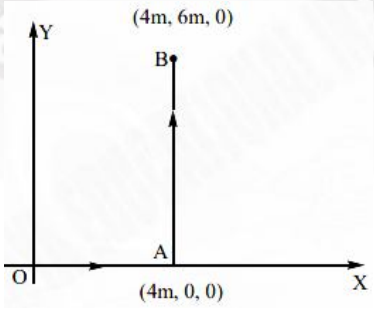
16. 3.2g of oxygen starts from the initial state $1(p_1 = 1.0\,atm, V_1 = 1.0L)$, reaches state 2 $(V_2 = 2.0L)$ through an isobaric process, and then passes through the isochoric process. The process reaches state $3(p_3 = 2.0atm)$, and then reaches state 4 through an adiabatic process. The temperature of state 4 is exactly the same as the temperature of state 1. Finally. It return to state 1 through an isothermal process, forming a cyclic process. $(Take\, P_4 = 0.0157\,atm, V_4 = 64L)$ The efficiency of cyclic process is $1-\frac{3x\ln 2}{17}$ then x value is



17. A charge q is surrounded by a closed surface consisting of an inverted cone of height h and base radius R, and a hemisphere of radius R as shown in the figure. The electric flux through the conical surface is $\frac{nq}{6\varepsilon_0}$ (in SI units). The value of n is _____.



18. Find the work done (in Joule) by a force, $\vec{F} = \left[\left(y^2 - x^3 + 5z^2\right)\hat{i} + (3xy - 2z)\,\hat{j} + (3z)\hat{k}\right]$ newton in taking a particle from origin O to the point 'B' (4m, 6m, 0) along the path OAB as shown in the figure



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CHEMISTRY **Max Marks: 62**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct. **Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)**

- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If the answer is correct)
  Zero Marks: **0** (If the question is unanswered)
  Negative Marks: -1 (In all other cases)

19. At same temperature and pressure a certain gas effuses out through two different vessels A and B separately. A has a circular orifice of radius is equal to twice of the side length of square orifice of vessel B. The ratio of rate of effusion of the gas from vessel A to that from vessel B is

A) $\pi : 4$ **B)** $4\pi : 1$ **C)** $4 : \pi$ **D)** $\pi : 1$

20. The crystal AB (rock salt structure) has molecular mass 6.023 y amu, where y is an arbitrary number in amu. If the minimum distance between cation and anion is $y^{1/3}$ nm then the density of AB(s) in kg / m^3 is

A) 20.0 **B)** 5.0 **C)** 10.0 **D)** 2.5

21. Which of the following statement is incorrect?

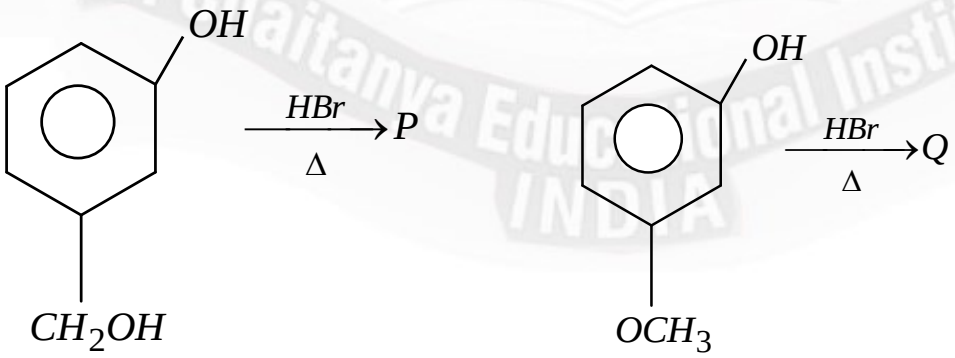
A) Tetracyanocuprate (I) is a tetrahedral complex

B) Tetracyanonickelate (II) is a square planar complex

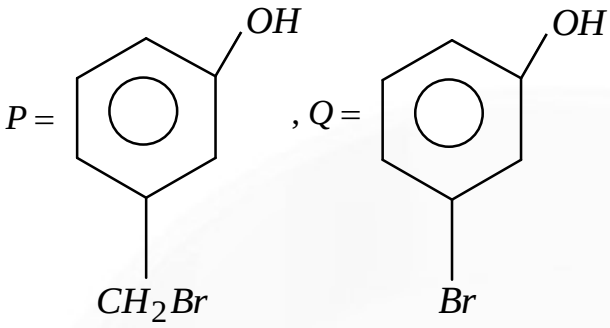
C) Tetrachloridoplatinate (II) is a tetrahedral complex

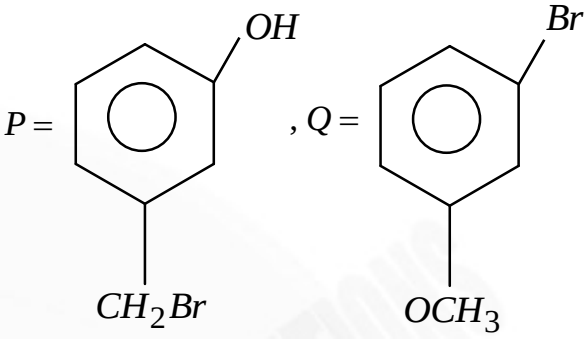
D) Tetraammine copper (II) sulphate is a square planar complex

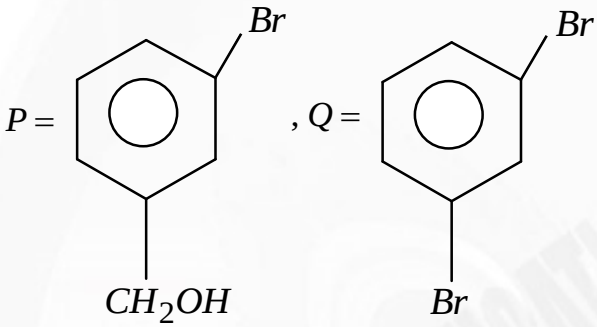
22. 'P' and 'Q' formed in the following set of reactions are

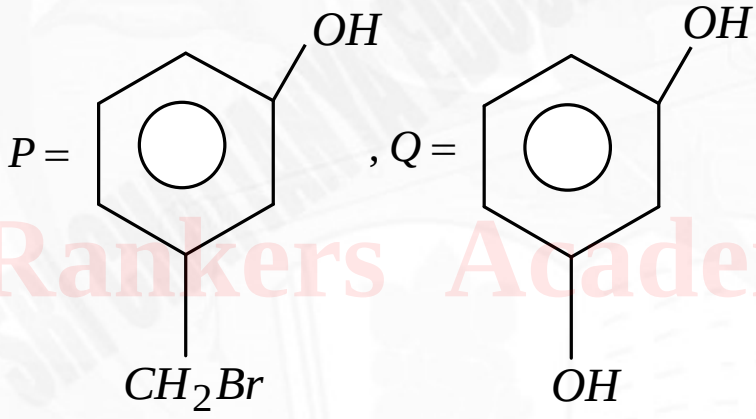


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THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023
JEE MAIN 2023 RANK 1
JEE Advanced 2023 RANK 1
NEET 2023 RANK 1
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A) 

B) 

C) 

D) 

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**

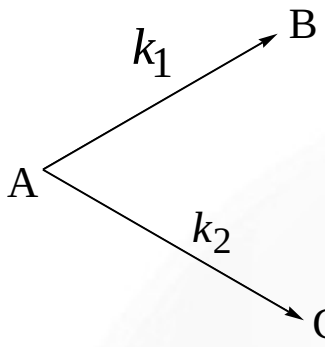
Section 2 (Max Marks: 32)

- Section two contains Eight (8) questions
- Each Question has Four Options and One Or More Than One of these four will be the correct answer.
- For each question, choose the option(s) corresponding to all the correct answer(s)

The Marking scheme to evaluate Answer to each question will be :
Full Marks: **+4** (If the answer is correct)
Partial Marks: **+3** (If all the four options are correct but **only 3** are chosen)
Partial Marks: **+2** (If THREE or More options are correct but **only 2** correct answers are chosen)
Partial Marks: **+1** (If TWO or More options are correct but **only 1** correct answer is chosen)
Zero Marks: **0** (If the question is unanswered)
Negative Marks: -1 (In all other cases)

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23. For the first order parallel reactions starting with 1 M of A,



Where k_1 *and* k_2 are rate constant for the formation of B and C respectivley the concentration of A as a function of time is given by:

$$[A](t) = [A]_0 e^{-(k_1+k_2)t}$$

The concentration of B and C will be

$$[B](t) = [A]_0\left(1-e^{-k_1 t}\right)$$

$$[C](t) = [A]_0\left(1-e^{-k_2 t}\right)$$

Select the correct statements for the given first order parallel kinetics

A) The concentration of A decreases exponentially with time according to the rate constant $k_1 - k_2$

B) The concentration of B increases according to the rate constant k_1 and follows 1st order kinetics.

C) The sum of the rates of formation of B and C is equal to the rate of disappearance of A

D) The concentration of A will never zero, but will approaches to zero as time increases.

24. The variation of EMF of a cell with temperature T(K) for the cell reaction

$$2\,Ag(s) + Hg_2Br_2(s) \rightarrow 2AgBr(s) + 2Hg(l)$$

Temperature in Kelvin	EMF of the cell (in mv)
293 K	66.3 mv
298 K	68.4 mv
303 K	70.5 mv

Which of the following statements is/are correct for the given cell reaction at 300 K

Assume 1Farday is equal to 96000 Columbs

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A) Emf of the cell at 300K is nearly equal to 69.24 mv

B) Entropy change of the given cell reaction at 300K is nearly equal to 80.64 $J\,K^{-1}$

C) Enthalpy change of the given cell reaction at 300 K is nearly equal to 10.90 KJ

D) Gibbs free energy change of the given cell reaction at 300 K is nearly equal to 13.29 KJ

25. A 0.252 g solid sample of unknown organic base is dissolved in 40 mL water and titrated with 0.14 M HCl solution. After the addition of 20 mL of acid, a pH of 10.7 is recorded. The equivalence point is reached when a total of 40 mL of HCl is added. The base and acid combine in a 1 :1 mole ratio then Which of the following statements is/are correct? (assume log 3 = 0.48, log 7 = 0.85)

A) The molar mass of the organic base is 45

B) K_b value of organic base is 5×10^{-4}

C) The pH of result mixture when the organic base is neutralized by 30 mL of HCl acid is 10.22

D) At equivalence point the pH of resulting mixture is 5.925

26. Select the correct order of dipole moment is?

A) $CH_3Cl > CH_3F$ **B)** $NO > CO$

C) $SO_2 > CO_2$ **D)** $SO_3 > SO_2$

27. Which of the following Reactions are/is not an example of Redox Reaction?

A) Hydrolysis of XeF_2 **B)** Hydrolysis of XeF_6

C) Reaction of XeF_4 *with* O_2F_2 **D)** Reaction of XeF_2 *with* PF_5

28. Choose the correct options according to MOT?

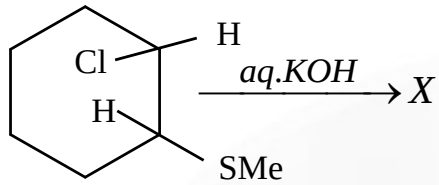
A) Bond order for C_2 molecule is 2 **B)** Molecular orbitals are polycentric

C) Bond length order : $B_2 > F_2$ **D)** Bond energy order : $B_2 > F_2$

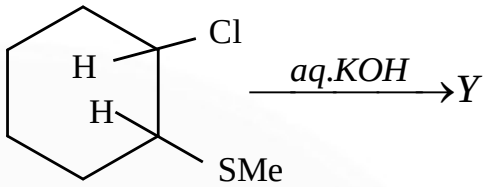
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29.

I)



II)



Select incorrect statement for above reaction:

A) X & Y are same compound, reaction I & II follow same mechanism

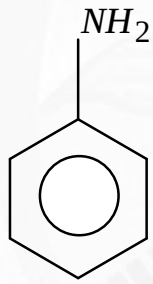
B) X & Y are same compound, reaction I &II follow different mechanism

C) X & Y are different compound, reaction I & II follow same mechanism

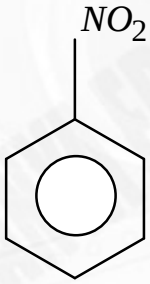
D) X & Y are different compound, reaction I &II follow different mechanism

30. Which of the following compound(s) cannot give friedel crafts reaction?

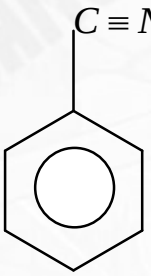
A)



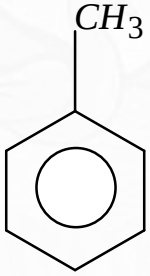
B)



C)



D)



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SECTION – III
(NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. **(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).**
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 3 (Max Marks: 18)

- This Section contains **Six (06)** Questions. The answer to each question is a Numerical Value
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If ONLY the correct numerical value is entered)
  Zero Marks: 0 (In all other cases)

31. The emf of the cell, $Pt(s), H_2(g,1bar)\left|H^+(Buffer)\right\|Hg_2Cl_2(s),KCl(1M)\left|Hg(l),Pt(s)\right.$ is 0.58 V at 298 K.

If $E^0_{cell}=0.28\,V, \dfrac{2.303\,RT}{F}=0.06V$ the pH of the buffer is …….

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32. An electron, initially at rest, is accelerated through a potential difference of 100 volts. It acquires debroglie wavelength $\lambda_1 \overset{o}{A}$. It is then get retarded successively by potentials 19 volts and 32 volts then, acquiring debroglie wavelengths $\lambda_2 \overset{o}{A}$. And $\lambda_3 \overset{o}{A}$ respectively. If the value of $\frac{\lambda_3 - \lambda_2}{\lambda_1} = \frac{x}{y}$ where x and y are simplest integers, then the magnitude of difference between x and y is

33. How many of the following hydroxides are soluble in excess NaOH solution?
$Fe(OH)_3, Al(OH)_3, Zn(OH)_2, Ni(OH)_2, Mn(OH)_2, Sn(OH)_2, Cu(OH)_2$

34. $Cl_2 + F_2(excess) \xrightarrow{573K} A \xrightarrow{Uranium} B + C$ [C is inter halogen compound]
The total No.of L.P's in A *is* $\rightarrow x$
The total No.of L.P's in B *is* $\rightarrow y$
The total No.of L.P's in C *is* $\rightarrow z$
$x + y - z = 15 + b$, Then 'b' is _____

35. The number of compounds among the following more reactive than acetic acid towards decarboxylation by soda-lime is
$O_2NCH_2COOH, HC \equiv CCH_2COOH, HC \equiv C - COOH, H_2C = CH - COOH,$
$PhCH_2COOH, Cl_3C - COOH, Me_3C - COOH,$
$CF_3COOH, CH_3CH_2COOH, Me_2CHCOOH, H_2C = CHCH_2COOH$

36. How many of the following can convert Ethyl magnesium bromide to Ethane?
$CH_2 = CH_2, CH_3CHO, HC \equiv CH, H_2O, CH_3CH_2OH, CH_3NH_2,$
$MeCOCH_2COMe, CH_3COCH_3, CH_3COOH, PhNH_2, CH_3C \equiv CH$

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MATHEMATICS **Max Marks: 62**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 4 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct. **Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases. Section 1 (Max Marks: 12)**

- Section 1 contains Four questions
- Each Question has Four Options and Only One of these four will be the correct answer.
- For each question, choose the option corresponding to the correct answer
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If the answer is correct)
  Zero Marks: **0** (If the question is unanswered)
  Negative Marks: -1 (In all other cases)

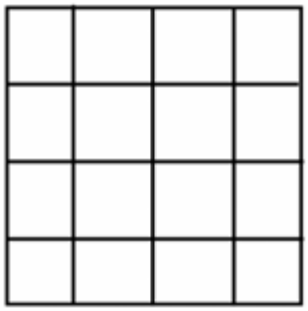
37. Consider two curves $C_1 : y^2 = 4\left[\sqrt{y}\right]x$ *and* $C_2 : x^2 = 4\left[\sqrt{x}\right]y$ *where* $[.]$ denotes the greatest integer function), then the area of region enclosed by these two curves within the square formed by the lines $x=1, y=1, x=4, y=4$ is

A) $\frac{11}{9}$ **B)** $\frac{11}{7}$ **C)** $\frac{11}{5}$ **D)** $\frac{11}{3}$

38. Let $f(x) = ax^2 + bx + c,\ a \neq 0,\ a,b,c \in I$. Suppose that $f(1) = 0,\ 50 < f(7) < 60$ *and* $70 < f(8) < 80$. If $13k < f(10) < 13(k+1)$ for some integer k, then value of k is

A) 5 **B)** 8 **C)** 10 **D)** 12

39. Four letters, two 'a' and two 'b' are filled into 16 cells of a matrix as given. It is required that each cell contains atmost one letter and each row or column cannot contain same letters. Then the number of ways the matrix can be filled is



A) 3600 **B)** 5200 **C)** 3960 **D)** 4120

40. If equation $4x^4 - ax^3 + bx^2 - cx + 6 = 0$ has 4 distinct real roots say $x_1 < x_2 < x_3 < x_4$ such that $\frac{1}{x_1} + \frac{2}{x_2} + \frac{3}{x_3} + \frac{4}{x_4} = 8;\ a,b,c \in R^+, then \frac{x_4}{x_1}$ is equal to

A) 2 **B)** 3 **C)** 4 **D)** 6

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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct. **Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -1 in all wrong cases**

Section 2 (Max Marks: 32)

- Section two contains Eight (8) questions
- Each Question has Four Options and One Or More Than One of these four will be the correct answer.
- For each question, choose the option(s) corresponding to all the correct answer(s)

The Marking scheme to evaluate Answer to each question will be :
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Partial Marks: **+1** (If TWO or More options are correct but **only 1** correct answer is chosen)
Zero Marks: **0** (If the question is unanswered)
Negative Marks: -1 (In all other cases)

41. If the conics whose equations are

$$S_1:\left(\sin^2\theta\right)x^2+\left(2h\tan\theta\right)xy+\left(\cos^2\theta\right)y^2+32x+16y+19=0$$

$$S_2:\left(\cos^2\theta\right)x^2-\left(2h'\cot\theta\right)xy+\left(\sin^2\theta\right)y^2+16x+32y+19=0$$

intersect in four noncyclic points, where $\theta\in\left[0,\frac{\pi}{2}\right]$, then the correct statement(s) can be

A) $h+h'=0$ **B)** $h-h'=0$ **C)** $\theta=\frac{\pi}{4}$ **D)** $\theta=\frac{\pi}{3}$

42. Consider the ellipse $\frac{x^2}{3969}+\frac{y^2}{256}=1$

Then which of the following statements is/are TRUE?

A) The least length of the segment of a tangent to the ellipse intercepted between the coordinate axes is 79

B) The least length of the segment of a tangent to the ellipse intercepted between the Coordinate axes is 65

C) The greatest distance of the centre of the ellipse from a normal to it is 47

D) The greatest distance of the centre of the ellipse from a normal to it is 65

43. If for non-zero complex numbers z_1 *and* z_2, $|z_2|\neq 1$, $z_1-iz_2=|z_2|z_1+i|z_1|z_2$, then which of the following statements is/are TRUE?

A) $z_1\overline{z_2}+\overline{z}_1z_2=0$ B) $z_1\overline{z_2}-\overline{z}_1z_2=0$

C) $z_1=\lambda z_2,\ \lambda\in R$ D) A possible value of $\arg\left(\frac{z_1}{z_2}\right)$ *is* $\frac{\pi}{2}$

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44. Let α, β, γ be the parametric angles of three points P, Q and R respectively lying on the circle $x^2 + y^2 = 1$. If the lengths of chords AP, AQ and AR are in G.P. where A is (1, 0) then $(let\ 0 \le \alpha, \beta, \gamma \le 2\pi)$

A) $\sin\frac{\alpha+\gamma}{4}\cos\frac{\alpha-\gamma}{4} \ge \sin\frac{\beta}{2}$ **B)** $\sin\frac{\alpha+\gamma}{4}\cos\frac{\alpha-\gamma}{4} \le \sin\frac{\beta}{2}$

C) $\sin\frac{\alpha}{2}\sin\frac{\gamma}{2} \ge \sin\frac{\beta}{2}$ **D)** $\sin\frac{\alpha}{2}\sin\frac{\gamma}{2} \le \sin\frac{\beta}{2}$

45. Consider the equation in real number x

$(x+1)^5 = \lambda\left(x^5+1\right), \quad \lambda \in R$

Then which of the following statements is/are TRUE?

A) For $\lambda = 2$, the equation has 3 distinct real solutions

B) For $\lambda = \frac{22}{7}$, the equation has 2 distinct real solutions

C) For $\lambda = \pi + \frac{22}{7}$, the equation has 3 distinct real solutions

D) For $\lambda = e$, the equation has 2 distinct real solutions

46. If the equation of pair of asymptote of a hyperbola H is $xy - 2x - y + 2 = 0$. Its conjugate hyperbola H' passes through (3, 0) then

A) eccentricity of H' is $\sqrt{2}$

B) focus of H' is $\left(2+\frac{1}{\sqrt{2}}, 2+\frac{1}{\sqrt{2}}\right)$

C) focus of H is $\left(\sqrt{2}, -\sqrt{2}\right)$

D) equation of transverse axis of H is $x - y + 1 = 0$

47. Let $X = \sum_{r=0}^{n}\left(\frac{(-1)^r}{r+1}\right)$, then $|X - \ln 2|$ will be

A) less than $\frac{2}{(n+1)}$ B) greater than $\frac{1}{2(n+2)}$

C) less than $\frac{1}{(n+1)}$ D) greater than $\frac{1}{4(n+2)}$

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48. If for $a,b>0$, $a.2^x+b.3^y+1-\sqrt{4^x+9^y+1}\sqrt{a^2+b^2+1}\geq 0$ holds true for some real x and y, then

A) A.M of a, b is $\frac{2^x+3^y}{2}$

B) $x=9^{2009}, y=(2009)^9$

C) A.M of a, b is $\frac{4^x+9^y}{2}$

D) $x\log_a 2+y\log_b 3=2$

SECTION – III
(NUMERICAL VALUE TYPE)

This section contains 6 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. **(e.g. 003.2343, 6.250023, 7.000000, -0.33, -.30, 30.27000, -127.0030).**
Marking scheme: +3 for correct answer, 0 if not attempted and 0 in all other cases. Section 3 (Max Marks: 18)

- This Section contains **Six (06)** Questions. The answer to each question is a Numerical Value
- The Marking scheme to evaluate Answer to each question will be :
  Full Marks: **+3** (If ONLY the correct numerical value is entered)
  Zero Marks: 0 (In all other cases)

49. If $a=\frac{2\pi}{1999}$ *and* $\cos a.\cos 2a.\cos 3a.....\cos 999a=\frac{1}{k}$, *then* $\log_2 k=$_____

50. If k be the value of x at which the function $f(x)=\int_{-1}^{x} t\left(e^t-1\right)(t-1)(t-2)^3(t-3)^5\,dt$ has local maximum value and $\sin x+\operatorname{cosec} x=k$, then for $n\in N$, $\sin^n x+\operatorname{cosec}^n x=$_____-

51. If $P\equiv\left(1,7,\sqrt{2}\right)$ be a point and line L is $2\sqrt{2}(x-1)=y-2, z=0$. If PQ is the distance of plane $\sqrt{2}x+y-z=1$ from point P measured along a line inclined at an angle of 45^0 with the line L and is minimum. The value of PQ is _____

52. Let A and B be two non-singular square matrices such that $B\neq I$ *and* $AB^2=BA$. *If* $A^3=B^{-1}A^3B^n$, then value of n is _____

53. From any point on the line $(t+2)(x+y)=1, t\neq -2,$ tangents are drawn to the ellipse $4x^2+16y^2=1$. It is given that chord of contact passes through a fixed point. Then the number of integral values of 't' for which the fixed point always lies inside the ellipse, is _____

54. If $f(x)=\sin^3 x+8+p^3-6p\sin x$, *where* $p>0$, then number of solutions of $f(x)=0$ for all p is equal to _____

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4 Students in Top 11 in JEE-Advanced 2025, All India Open Category

Rankers Academy JEE

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30 RANK OM PRAKASH BEHERA HT. No. 252021018*	32 RANK SUNKARA SAI RISHANTH REDDY HT. No. 256165327	34 RANK DHRUBA JYOTHI PANJA HT. No. 252048248*	35 RANK BHAVESH JAYANTHI HT. No. 251043080	36 RANK ADVAY MAYANK HT. No. 252104113*	37 RANK KARMANYA GUPTA HT. No. 252081477*
42 RANK MD ANAS HT. No. 252046210*	45 RANK RAMIT GOYAL HT. No. 257001113*	52 RANK MAULIK JAIN HT. No. 252079407*	54 RANK GARV HT. No. 252056188*	59 RANK LARISSA HT. No. 252079071*	60 RANK ARYAN BALABADRULA HT. No. 256132077
63 RANK SAMYAJYOTI BISWAS HT. No. 255058456*	64 RANK AARUSH ANAND HT. No. 251006176*	72 RANK RUSHMITH BANDARI HT. No. 256168048	78 RANK KORIKANA RASAGNYA HT. No. 256057046	87 RANK LAKSHYA SHARMA HT. No. 252070079*	91 RANK AVANEESH BANSAL HT. No. 251113130*	95 RANK KAVYA AGGARWAL HT. No. 252079121*

BELOW 100 ALL INDIA OPEN CATEGORY RANKS: 29	BELOW 500 ALL INDIA OPEN CATEGORY RANKS: 113	BELOW 1000 ALL INDIA OPEN CATEGORY RANKS: 205	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT: 745	NUMBER OF QUALIFIED RANKS: 4,212

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